

CAN WE PLAN MEGA EVENTS USING AI?

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Introduction

Mega events such as the Olympic Games, the FIFA World Cup, and the Hajj pilgrimage pose unique challenges due to their massive scale, logistical complexity, and societal impact. These events require coordination across infrastructure, security, transportation, and sustainability. Traditional planning methods, often reactive and manual, struggle to address such dynamic, high-stakes conditions. Artificial intelligence (AI) has emerged as a pivotal innovation, enabling predictive, real-time, and data-driven decision-making. Evidence from Paris 2024 shows how digital twins simulate venue layouts, optimize crowd flows, and reduce coordination costs. In the Hajj, AI applications have enhanced mobility management and health monitoring for millions of pilgrims. This study investigates the transformative potential of AI in mega event management through a systematic synthesis of recent scholarship and real-world case studies.

Objectives

- Identify scalable AI applications in mega event planning and management.
- Assess ethical, governance, and scale-specific challenges in AI integration.
- Foster managerial adoption by demonstrating efficiency, risk mitigation, and stakeholder satisfaction.

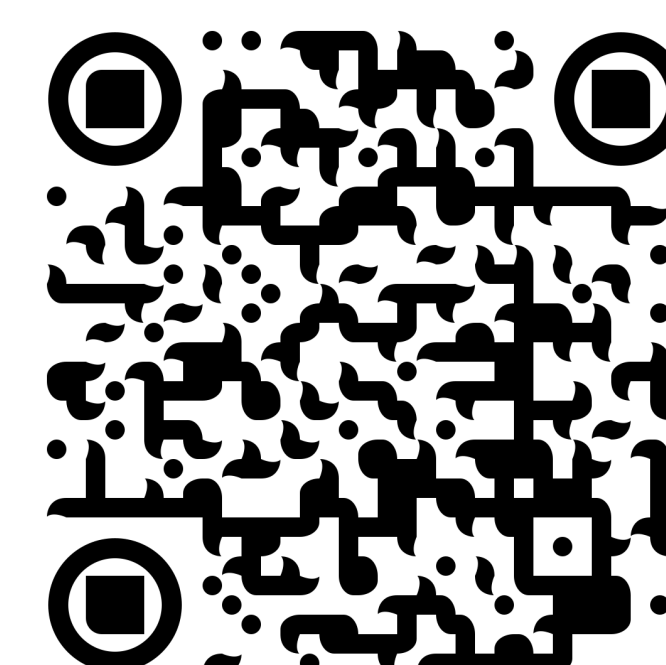
Literature Review

The integration of AI in event management has evolved from conceptual exploration to practical implementation. AI in Crowd and Risk Management: Machine learning improved flow efficiency by 30% in Hajj. Digital Twins and Simulation: Perception-powered digital twins enabled virtual testing of 15 million visitor scenarios in Paris 2024. Generative AI and Automation: Tools like ChatGPT automated scheduling and marketing, achieving 40% efficiency gains. Ethical and Governance Dimensions: Privacy risks in AI surveillance require ethical leadership. Despite benefits, managerial resistance due to distrust persists.

AI Application Framework for Mega Events



References available upon request



Methodology

This research adopts a two-phase design: a Systematic Literature Review (SLR) following PRISMA guidelines and a multiple case study analysis. The SLR will search major academic databases and peer-reviewed publications from 2019–2025. Keywords include “mega event,” “Olympics,” “FIFA World Cup,” and “Hajj” combined with “artificial intelligence,” “planning,” and “management.” Twenty-five high-quality sources have been preliminarily identified and will be evaluated using the Mixed Methods Appraisal Tool (MMAT). The case studies analyze the Paris 2024 Olympics (digital twins for venue simulation), the FIFA World Cup Qatar 2022 (AI for crowd & transport management), and the Hajj 2024–2025 (real-time health & density forecasting), emphasizing scalable applications, governance challenges, and ethical implications.

Expected Results/Implications

The synthesis of the literature and case studies reveals four key AI application themes that directly support the research objectives:

- (1) Predictive logistics for optimized transport and crowd flow enhances operational efficiency and risk mitigation
- (2) Generative content for automated scheduling and marketing reduces administrative burden and improves stakeholder engagement
- (3) Real time governance systems for health, safety, and sustainability addresses scale specific and ethical challenges
- (4) Post event learning via digital twin feedback loops facilitates knowledge transfer and builds long term managerial confidence.

By empirically illustrating these tangible benefits through real world mega event cases, the study provides concrete evidence to overcome managerial distrust and resistance to AI adoption. Theoretically, it bridges event management with AI ethics; practically, it offers a conceptual framework to guide ethical, transparent, and trust building AI integration, ultimately supporting more resilient and sustainable mega events for future hosts.